

Reading a connectome out of a graph network: the template readout and the gauge it has to undo

Abstract

A graph network trained to predict neural dynamics can reproduce every trajectory while getting every synapse wrong, because only the sum of a neuron’s own dynamics and its incoming message is observed. This note writes down the transformation that freedom permits — it is affine, not a pure scale — and gives a closed-form readout that inverts it: fit the generator’s own synaptic form to the model’s per-edge message, then carry the result into the generator’s units with a factor measured from the model’s update. Every step is a least squares with an explicit solution, so the whole 434,112-edge connectome is read in seconds. On a conductance network trained at process noise $\sigma = 0.05$ the readout recovers the median synaptic conductance to 7.3% and raises cell-type clustering accuracy from 0.490 to 0.583; it also shows, and quantifies, the two failures that remain.

1 What is observed, and what is free

The generator is a conductance-based network of N neurons. For neuron i ,

$$\tau_i \frac{dv_i}{dt} = -(v_i - V_i^{\text{rest}}) + \text{msg}_i + I_i, \quad \text{msg}_i = \sum_{j \in \mathcal{N}(i)} W_{ij} \text{act}(v_j) (E_{ij} - v_i), \quad (1)$$

with $\text{act} = \text{relu}$ on this data, $W_{ij} \geq 0$ a synaptic conductance in units of the leak conductance, and E_{ij} the reversal potential of the channel the presynaptic cell opens. The sign of a synapse lives entirely in the driving force $E_{ij} - v_i$, never in W_{ij} .

The model is a graph network with the same connectivity:

$$\frac{dv_i}{dt} = f_\theta(v_i, a_i, \widehat{\text{msg}}_i, I_i), \quad \widehat{\text{msg}}_i = \sum_{j \in \mathcal{N}(i)} \widehat{W}_{ij} g_\phi(v_j, a_j, v_i, a_i), \quad (2)$$

where a_i is a learned per-neuron embedding and $\widehat{W}_{ij}$ a learned per-edge weight. Only dv_i/dt is supervised. *Nothing in the loss observes $\widehat{\text{msg}}$* , and that is the whole difficulty: the message is an internal variable, and any transformation of it that f_θ can undo leaves the training objective exactly unchanged.

1.1 The gauge is affine

Suppose the model’s message is related to the truth by

$$\widehat{\text{msg}}_i = \frac{1}{k_i} \text{msg}_i + \beta_i. \quad (3)$$

Write the model’s update in the same form as the generator’s, which is what the update fit of §2.2 measures:

$$\frac{dv_i}{dt} = T_i \left[(V_i - v_i) + G_i \widehat{\text{msg}}_i + f(I_i) \right]. \quad (4)$$

Substituting (3) into (4) and matching coefficients against (1) term by term gives three identities:

$$\text{message:} \quad T_i G_i \frac{1}{k_i} = \frac{1}{\tau_i} \implies k_i = \tau_i T_i G_i, \quad (5)$$

$$\text{leak:} \quad T_i = \frac{1}{\tau_i} \implies \tau_i = 1/T_i, \quad (6)$$

$$\text{constant:} \quad T_i V_i + T_i G_i \beta_i = \frac{V_i^{\text{rest}}}{\tau_i} \implies V_i^{\text{rest}} = \tau_i T_i V_i + k_i \beta_i. \quad (7)$$

Three consequences, and they are the reason this note exists.

1. **The scale k_i is unobservable.** Equation (5) is satisfied for *any* k_i , by pairing it with $G_i = k_i/(\tau_i T_i)$. A model whose messages are $25\times$ too large and whose f_θ divides by 25 predicts precisely the same dynamics as the truth.
2. **The offset β_i is unobservable too, and it is not harmless:** by (7) it is absorbed into the resting potential. A constant in the message and a shift of V^{rest} are the same function.
3. **The reversal E_{ij} is gauge-invariant.** It is the voltage at which a synapse’s message vanishes — a root — and a root does not move when the function around it is rescaled or, per edge, shifted by a constant that the fit accounts for separately. This is why E_{ij} can be recovered absolutely while W_{ij} cannot be, without further work.

2 The readout

The readout inverts (3) in three stages: fit the generator’s synaptic form to each edge (§2.1), fit the generator’s update form to each neuron (§2.2), and combine them (§2.3).

2.1 Stage 1: the per-edge template

For every edge (i, j) and every sampled frame t we have the model’s own per-edge message, which is (2) before the sum,

$$m_{ij}(t) = \widehat{W}_{ij} g_\phi(v_j(t), a_j, v_i(t), a_i). \quad (8)$$

We ask what conductance, reversal and offset best explain it inside the generator’s own form. Writing $u_{ij}(t) = \text{act}(v_j(t))$ — the generator’s activation, *given*, not fitted —

$$m_{ij} \approx W_{ij} u_{ij} (E_{ij} - v_i) + C_{ij} = \underbrace{W_{ij} E_{ij}}_{b_1} u_{ij} + \underbrace{(-W_{ij})}_{b_2} u_{ij} v_i + \underbrace{C_{ij}}_{b_3} \cdot 1. \quad (9)$$

With act given, (9) is *linear in its three unknowns*: a three-column least squares with a closed-form solution, one per edge, solved for all edges at once from stacked normal equations. This is the step that would otherwise be a symbolic regression — searching for act as well costs about 300s per *neuron*, against seconds for the entire connectome here.

Three details matter.

The offset C_{ij} is a measurement, not a nuisance. The generator’s per-edge message is identically zero whenever the presynaptic cell is silent, for every edge and whatever the postsynaptic state; g_ϕ is under no such obligation. C_{ij} is exactly what it emits there, and by (7) the incoming sum $\sum_j C_{ij}$ is what neuron i ’s resting potential absorbs. Fitting it also unbiases the other two coefficients: without a constant in the model, a message sitting on a pedestal is fitted by tilting the driving force instead, which moves the reversal.

Frames with no drive carry no information. Only frames with u_{ij} above the median of the positive activations enter the fit; below that the driving force is multiplied by almost nothing.

The reversal is reported only where it is identified. Since $E_{ij} = b_1/(-b_2)$ divides by the second coefficient, an edge whose postsynaptic voltage barely moved returns the reciprocal of a small noisy number. The edge reports a reversal only when

$$\frac{|b_2|}{\text{se}(b_2)} \geq t^*, \quad \text{se}(b_2)^2 = \frac{\text{SS}_{\text{res}}}{n-3} [(X^\top X)^{-1}]_{22}, \quad (10)$$

with $t^* = 3$. Before this gate, a handful of such edges drove the reversal RMSE to 571 V.

The conductance comes from the better-determined coefficient. The generator gives every edge onto neuron i one of exactly two reversals — one for the excitatory senders, one for the inhibitory — so the surviving per-edge estimates of each sign are repeated measurements of one number and their median $\bar{E}_i^\pm$ is the estimator that says so. Then

$$\widehat{W}_{ij}^{\text{fit}} = \frac{b_1}{\bar{E}_i^\pm} \quad \text{rather than} \quad \widehat{W}_{ij}^{\text{fit}} = -b_2, \quad (11)$$

because b_1 multiplies the drive itself while b_2 multiplies $u_{ij}v_i$ and is only as good as the postsynaptic voltage's excursion.

2.2 Stage 2: the per-neuron update

Equation (4) is likewise linear once $f(I)$ is taken linear:

$$\frac{dv_i}{dt} \approx a_0 + a_1 v_i + a_2 \widehat{\text{msg}}_i + a_3 I_i, \quad T_i = -a_1, \quad V_i = \frac{a_0}{T_i}, \quad G_i = \frac{a_2}{T_i}, \quad (12)$$

a four-column least squares per neuron, accumulated over frames as $(N, 4, 4)$ normal equations so that nothing of size (N, n_{frames}) is ever held. The quality of this linearisation is reported: on the example run the median per-neuron R^2 of (12) is 0.9994, and the fitted $T_i G_i$ agrees with an autograd evaluation of $\partial f_\theta / \partial \text{msg}$ to a median ratio of 0.9994 — the two coincide exactly when the update is affine in the message, so their ratio is a check that it is.

One confound must be named: a constant inside $f(I)$ reaches dv_i/dt as a constant times T_i , exactly as V_i does, so on neurons that receive a stimulus the two cannot be separated. On this data that is the photoreceptors only.

2.3 Stage 3: the gauge

From (5), each neuron's conductances are carried into the generator's units by

$$k_i = \tau_i T_i G_i, \quad W_{ij}^{\text{learned}} = k_i \widehat{W}_{ij}^{\text{fit}}, \quad E_{ij}^{\text{learned}} = E_{ij}^{\text{fit}}. \quad (13)$$

Which τ_i closes this conversion is a genuine methodological choice and the note takes a position on it.

- **Blind (default).** Use the model's own $\tau_i = 1/T_i$ from (6), whereupon $k_i = G_i$ exactly. Nothing from the generator enters, which is required of a recovery metric: τ is one of the quantities being recovered and cannot appear in its own readout.
- **Diagnostic.** Use the generator's τ_i . This is *not* a recovery number — it uses the answer — but it separates the template's own error from τ 's, because W^{true} is written in units where the message enters dv_i/dt as $\text{msg}/\tau^{\text{true}}$.

The difference between them is large and is itself a result: see §4.

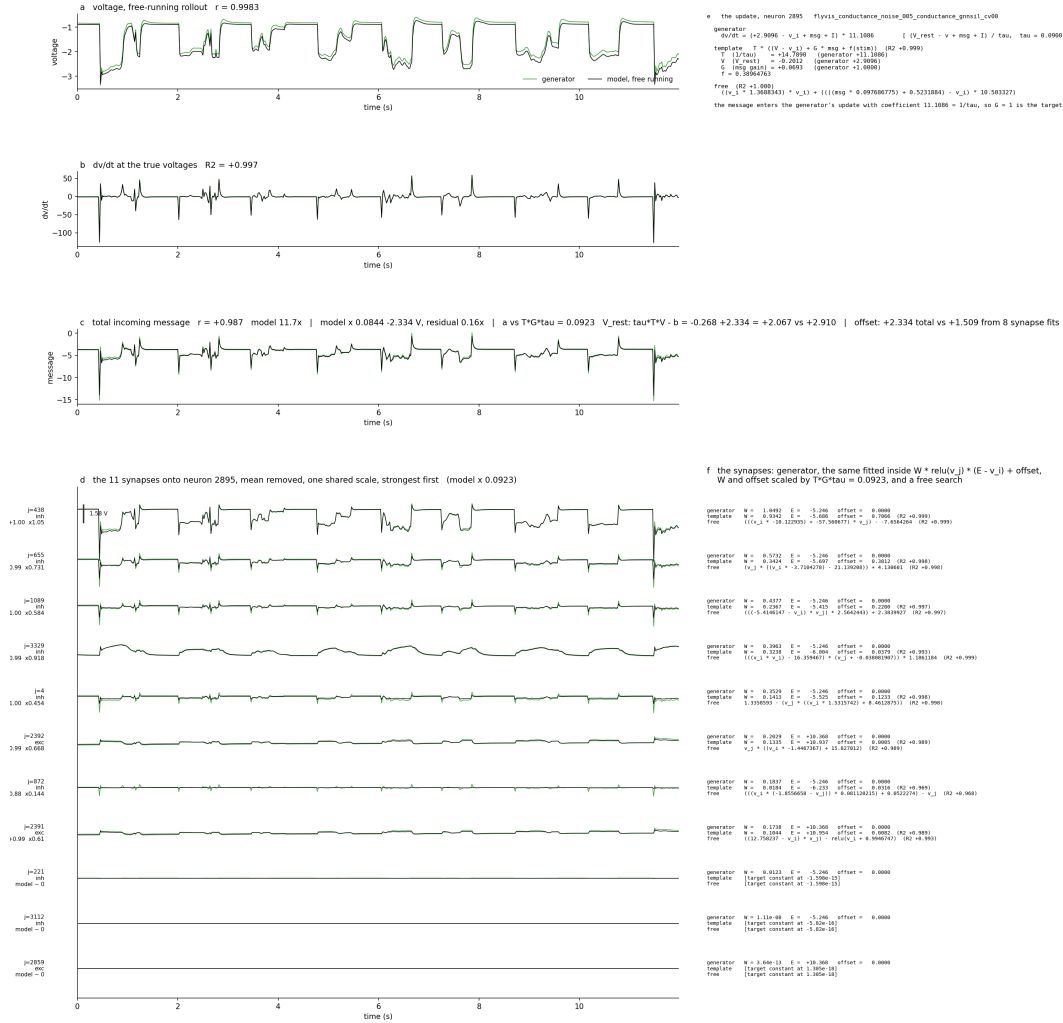


Figure 1: One neuron, term by term. **(a)** free-running rollout, generator in green and model in black, $r = 0.998$. **(b)** dv/dt at the true voltages, $R^2 = 0.997$. **(c)** the total incoming message: the learned one after the affine correction of (3), against the generator's. The heading reports the fitted scale 0.0844 against $TG\tau = 0.0923$ from panel (e) — two independent routes to the same number, agreeing to 9% — and the offset $b = -2.334 \text{ V}$. **(d)** each of the 11 synapses separately, mean removed, on one shared voltage scale so that a synapse the model dropped looks flat rather than being magnified to full height. **(e)** the update fit of (12): $T = 14.79$ against the generator's $1/\tau = 11.11$, $G = 0.0693$ against a target of 1. **(f)** per synapse, the generator's ($W, E, \text{offset} = 0$) against the template fit of (9), and a free symbolic search for comparison.

3 A worked example

Figure 1 shows the readout applied to neuron 2895 of a conductance network trained at process noise $\sigma = 0.05$ on the flyvis connectome ($N = 13,741$ neurons, 434,112 edges).

The V^{rest} identity closes. Equation (7) predicts $V^{\text{rest}} = \tau TV - b$ where $b = -k\beta$ is the fitted offset of panel (c). Substituting: $-0.083 + 2.334 = +2.251$ against the generator’s $+2.910$ — 77% of the resting potential recovered, of which essentially all comes from the message’s offset and almost none from the update’s own level $V = -0.21$. Reading V alone as V^{rest} , as a naive fit of (4) invites, would be wrong by 3 V.

The per-edge offsets sum to the per-neuron one. Panel (c) reports $+2.334$ measured on the total message against $+1.510$ from summing the eight fitted C_{ij} ; the gap is the three synapses the model pruned, which have no fit to contribute one. The two stages agree on a quantity neither of them imposes.

4 Population results

Table 1: The same trained model, read three ways. “Chain” is the previous estimator (a per-edge line through m/v_j with the g_ϕ and f_θ gain corrections). “Blind” and “diagnostic” are the two gauge conventions of §2.3. W statistics are over the 321,774 edges the fit accepted; R^2 is outlier-filtered (383 outliers, 0.12%).

	chain	template, blind	template, diagnostic
$W_{ij} R^2$	—	+0.273	+0.691
W_{ij} Pearson r	—	0.238	0.796
W_{ij} median relative error	—	25.2%	7.3%
W_{ij} slope	—	0.582	0.760
E_{ij} median per-edge fit R^2	0.812	0.9997	0.9997
E_{ij} RMSE (V)	—	7.76	7.76
τR^2	0.912	0.925	0.925
$V^{\text{rest}} R^2$	0.787	0.792	0.792
clustering accuracy	0.490	0.583	0.662
clustering NMI	0.714	0.828	0.838

Three readings of Table 1 deserve emphasis.

The form is identified; the constant inside it is not. The median per-edge R^2 of the conductance template is 0.9997: the model’s message *is* of the form $W^{\text{act}}(v_j)(E - v_i) + C$ to four decimal places. The reversal implied by that form nonetheless has an RMSE of 7.76 V across the 66 distinct values the generator uses. Reporting the fit quality as if it were recovery would be reporting the extraction’s goodness of fit; both numbers belong in any claim about reversal potentials.

What separates blind from diagnostic is τ , and only τ . The two columns differ by nothing but which time constant closes (13). Since τ is recovered at $R^2 = 0.925$, one might expect the effect to be small; it is not, because k_i multiplies an entire row of the connectome, so each neuron’s τ error scales all of its synapses coherently. The median τ is 37% too large, and the W statistics inherit exactly that. *This makes a prediction that can falsify the whole framing:* a regulariser that pins $G_i \rightarrow 1$ forces $k_i = \tau_i T_i G_i \rightarrow 1$ and removes τ from the conductance readout altogether, so the two columns should converge. If they do not, the gauge is not the whole story.

The population number is dominated by synapses that do not exist. Figure 2 shows why the per-neuron picture (median learned/true 1.39, interquartile range 1.22) and the population R^2 (0.273) tell different stories. Half the fitted edges have a true conductance below 10^{-8} (the first five deciles run from 2×10^{-15} to 8.5×10^{-9}) — the generator has no synapse there — and the model gives them conductances up to 10^{-2} . They dominate the sum of squares while carrying no biology. A second, smaller population is whole postsynaptic neurons whose rows collapsed, visible as vertical streaks at ratios of 10^{-6} and below.

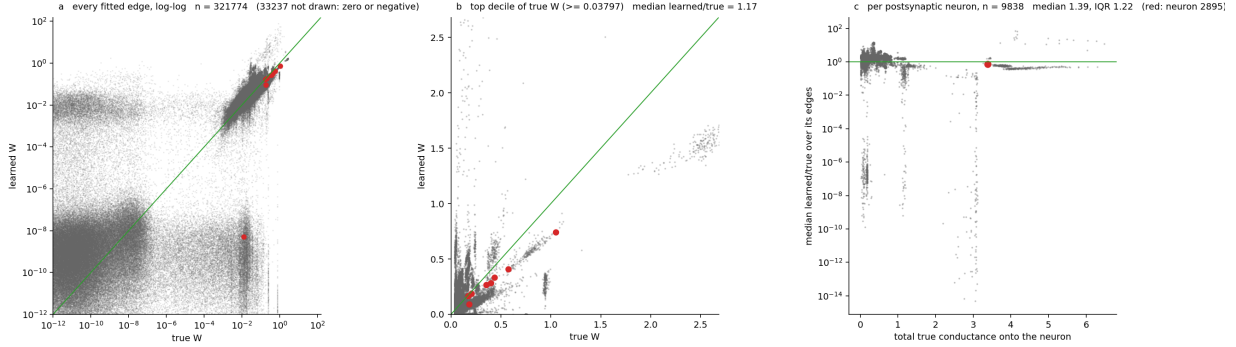


Figure 2: Learned against true conductance, blind gauge. **(a)** every fitted edge, logarithmic axes, with neuron 2895’s synapses in red; the diagonal ridge is the recovered population and the cloud at lower left is the phantom one. **(b)** the top decile of true conductance on linear axes, median learned/true 1.17. **(c)** per postsynaptic neuron, the median ratio over its edges against the total conductance it receives; the bulk sits at 1.39 and the vertical streaks are collapsed rows.

5 Is the method sound?

What it assumes. (i) That act is known. It is given from the generator, so on real data this becomes an assumption about the synaptic nonlinearity and should be stated as one; the free symbolic search in panel (f) of Figure 1 exists to check it, and a separate PySR template search — which fits act rather than assuming it — is run on a handful of neurons for exactly that purpose. (ii) That f_θ is approximately affine in the message over the observed range; measured, not assumed ($R^2 = 0.9994$, ratio to autograd 0.9994). (iii) That the generator gives each neuron two reversals, used only by the pooling step, which can be disabled.

What it does not do. It does not make an unidentifiable quantity identifiable. The offset β_i is measured here against the generator’s message, which is an *account* of the error, not a blind recovery; only a prior that pins the message’s level — a penalty on g_ϕ at silent presynaptic input — makes it identifiable during training. The same holds for the scale: the readout undoes k_i after the fact, and a penalty tying $\partial f_\theta / \partial \text{msg}$ to $-\partial f_\theta / \partial v$ is what would remove it during training.

What would falsify it. The convergence prediction of §4; the agreement of $\sum_j C_{ij}$ with the independently fitted total offset, which currently holds to the pruned synapses; and the agreement of the fitted $T_i G_i$ with autograd, which would break first if the update stopped being affine in the message.

6 Implementation

The readout is `metrics.extract_template_params`, scored through the same `score_recovery` as every other estimator so that the two readouts of a run differ only in `Wij_estimator` and can

be compared line by line. Per-edge fits are batched `numpy` solves of stacked 3×3 normal equations; the per-neuron update is a batched 4×4 solve accumulated over frames. Cost on the example run: all 434,112 edges and 13,741 neurons in under a minute on one L4, against the ~ 300 s per neuron a symbolic search over act would take. The figures in this note are generated by `GNN_Main.py -o plot` and `tools/wij_scatter.py`.